Math 340: Elementary Matrix and Linear Algebra

MIDTERM ONE

Wednesday, February 28th, 2024, 7:30pm-9:00pm

Please circle your Instructor and your Discussion Session (date and start time given) in the table below.

Dr. Péter Juhász	Dr. Sean Paul	Dr. Terry Harris
Awildo Gutierrez, M 7:45 AM	Jiaming Xu, T 11:00 AM	Ryan Tamura, T 12:05 PM
Awildo Gutierrez, M 8:50 AM	Jiaming Xu, T 12:05 PM	Rune Chen, T 1:20 PM
Hongyu Zhu, W 9:55 AM	Jiaming Xu, R 11:00 AM	Ryan Tamura, R 12:05 PM
Awildo Gutierrez, W 7:45 AM	Zaidan Wu, R 12:05 PM	Hanzhang Mao, R 1:20 PM
Hongyu Zhu, W 8:50 AM	Zaidan Wu, R 1:20 PM	Connor Simpson, T 2:25 PM
Jacob Denson, M 9:55 AM		Ryan Tamura, T 11:00 AM
Carsen Grote, M 8:50 AM		Lauren Neudorf, R 2:25 PM
Carsen Grote, M 9:55 AM		Connor Simpson, T 12:05 PM
Jacob Denson, W 8:50 AM		Hanzhang Mao, R 12:05 PM
Carsen Grote, W 9:55 AM		Connor Simpson, T 9:55 AM
Jacob Denson, W 9:55 AM		

Name: Vincent Adultman Wisc email: ______________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______________________________

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of seven problems for a total of 50 points; the exam is 6 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (5 points, 1 point each) Are each of the following true or false statements? Fill in the bubble next to the correct answer. Justification is not necessary.

a.) If A is a 10×20 matrix with $\text{rank}(A) = 10$, then $A\mathbf{x} = \mathbf{b}$ is consistent for any vector $\mathbf{b}$.

● True
○ False

b.) If A is a 3×3 matrix with $A^2 = A$, then $\det(A) = 1$.

○ True
● False

$\Downarrow$

$\det(A)^2 = \det(A)$

so $\det(A) = 0$ or 1

c.) If A is a 2×7 matrix and C is a 3×4 matrix, and ABC^T is a 2×3 matrix, then B must be 7×4.

● True
○ False

$2\times 7 \quad 4\times 3$

7×4

d.) If A is a 3×3 matrix with rank 2, then every linear system $A\mathbf{x} = \mathbf{b}$ will have infinitely many solutions.

○ True
● False

e.) If $T : \mathbb{R}^{\overset{4}{a}} \to \mathbb{R}^{\overset{2}{b}}$ is a matrix transformation defined by $T(\mathbf{x}) = A\mathbf{x}$ for A a 2×4 matrix, then $a = 4$ and $b = 2$.

● True
○ False

$2\times 4 \;\; 4\times 1 \to 2\times 1$

2. (7 points) Suppose A and B are 4×4 matrices with $\det(A) = 12$ and $\det(B) = 3$.

   a. (4 points) If C is a 4×4 matrix with $2A^{-1}B^TC + I_4 = O$ (where O is the 4×4 zero matrix), find $\det(C)$. Show all work.

$$2A^{-1}B^TC = -I_4$$

$\downarrow$ det

$$2^4 \frac{1}{\det(A)} \det(B) \det(C) = \det(-I_4) = 1.$$

$$2^4 \cdot \frac{1}{12} \cdot 3 \det(C) = 1$$

$$\det(C) = \frac{12}{2^4 \cdot 3} = \frac{1}{4}$$

   b. (3 points) Recall B is a 4×4 matrix with $\det(B) = 3$. Suppose D is a 4×4 matrix obtained from B by performing the following row operations on B:

      Swap r_2 and r_4 $\quad -3$
      Add $7r_1$ to r_2 $\quad -3$
      Multiply r_3 by $\frac{1}{6}$ $\quad -\frac{1}{6} \cdot 3$
      Add $-r_3$ to r_4 $\quad -\frac{1}{6} \cdot 3$
      Multiply r_2 by 10 $\quad 10 \cdot (-\frac{1}{6}) \cdot 3$

   Find $\det(D)$. Show all work.

$$\det(D) = 10(-\tfrac{1}{6})(3) = -5$$

3. (9 points total) (this problem has overflow room on the next page)

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ 2x_3 - x_1 \\ 3x_2 \end{bmatrix}$.

a. (2 points) Evaluate $T\left(\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}\right)$.

$$T\left(\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 2+1 \\ 2(4)-2 \\ 3(1) \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}$$

b. (3 points) Find the (standard) matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \qquad T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 2 \\ 0 & 3 & 0 \end{bmatrix}$$

c. (4 points) Is there an $\mathbf{x}$ in $\mathbb{R}^3$ such that $T(\mathbf{x}) = \begin{bmatrix} 1 \\ -3 \\ 6 \end{bmatrix}$? Fully justify your answer.

Yes.

① $x_1 + x_2 = 1 \Rightarrow x_1 = -1$

$2x_3 - x_1 = -3 \Rightarrow x_3 = \frac{-3-1}{2} = -2$

$3x_2 = 6 \Rightarrow x_2 = 2$

② $\det(A) = -3(0+1) = -3 \neq 0$, so $A\vec{x} = \vec{b}$ has a solution for every $\vec{b}$.

③ $$\begin{bmatrix} 1 & 1 & 0 & 1 \\ -1 & 0 & 2 & -3 \\ 0 & 3 & 0 & 6 \end{bmatrix} \xrightarrow{\cdots} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right]$$ yes!

(Overflow room for Problem 3, if needed)

4. (4 points) Find the eigenvalues of A, where $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$. Show all work.

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda-4 & -2 \\ 1 & \lambda-1 \end{bmatrix} = (\lambda-4)(\lambda-1)+2$$

$$= \lambda^2 - 5\lambda + 4 + 2$$

$$= \lambda^2 - 5\lambda + 6$$

$$= (\lambda-3)(\lambda-2)$$

$$\lambda = 3, 2$$

5. (8 points total) Suppose A and B are invertible 2×2 matrices with

$A^{-1} = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} -1 & 2 \\ 4 & 2 \end{bmatrix}$.

Use A^{-1} and B^{-1} to answer the following:

a. (4 points) Solve the linear system $C\mathbf{x} = \mathbf{b}$, where $C = AB$ and $\mathbf{b} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$. Show all work. (Hint: you do not need to find A and B to solve this problem.)

$$\vec{x} = C^{-1}\vec{b}, \quad C^{-1} = B^{-1}A^{-1} = \begin{bmatrix} -1 & 2 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 3 \\ 10 & 8 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} -5 & 3 \\ 10 & 8 \end{bmatrix}\begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 13 \\ -12 \end{bmatrix}$$

b. (4 points) Suppose D is a 2×2 matrix with $(3D + I_2)^{-1} = A$. Find D. Show all work.

$$3D + I_2 = A^{-1} = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$3D = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\boxed{D = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}}$$

6. (7 points) Suppose A is a 3×3 matrix, and $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ is an eigenvector of A with eigenvalue -3, and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$ is an eigenvector of A with eigenvalue 4.

Compute each of the following (Hint: Each of your final answers should be a vector in $\mathbb{R}^3$). Show all work.

a. (1 point)

$$A \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix} = -2A \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = -2 \begin{bmatrix} 8 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -16 \\ 0 \\ -8 \end{bmatrix}$$

b. (2 points)

$$A^2 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = A\,A \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = A(-3) \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = (-3)(-3) \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ -18 \\ 9 \end{bmatrix}$$

c. (4 points)

$$A \begin{bmatrix} 4 \\ -4 \\ 3 \end{bmatrix} = A\left(2\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\right) = 2A\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + A\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = -6\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 8 \\ 0 \\ 4 \end{bmatrix}$$

(Hint: Write $\begin{bmatrix} 4 \\ -4 \\ 3 \end{bmatrix}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$)

$$= \begin{bmatrix} 2 \\ 12 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ -4 \\ 3 \end{bmatrix} = a \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a+2b \\ -2a \\ a+b \end{bmatrix}$$

$-2a = -4$

$a = 2$

$a + b = 3$

$b = 1$

7. (10 points total) (this problem has overflow room on the next page)

Given $M = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -4 \\ 0 & 3 & -12 \end{bmatrix}$: $\begin{bmatrix} -6 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -6+8-2 \\ 4-4 \\ 12-12 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$!!

$-3r_2+r_1 \to r_1$

a. (4 points) Find the reduced row echelon form of M. Show all work.

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-2r_2+r_1 \to r_1} \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

b. (1 point) What is the rank of M?

2

c. (2 points) Write the matrix equation $M\mathbf{x} = \mathbf{0}$ as a system of three equations in three unknowns (x, y, z).

$$x + 2y - 2z = 0$$
$$y - 4z = 0$$
$$3y - 12z = 0$$

d. (3 points) Using your answer to part (a), solve the homogeneous linear system $M\mathbf{x} = \mathbf{0}$. Write your solution as a parameterized set (Hint: This system has infinitely many solutions. It's recommended to check your final answer is correct.).

z is free $\Rightarrow z = t$

$x + 6z = 0 \Rightarrow x = -6t$

$y - 4z = 0 \Rightarrow y = 4t$

Soln:

$$\begin{bmatrix} -6t \\ 4t \\ t \end{bmatrix} = t\begin{bmatrix} -6 \\ 4 \\ 1 \end{bmatrix}$$ check up above.

(Overflow room for Problem 7, if needed)

Scratch paper page!!

Back of scratch paper page!